

WASP-12b

R-1941

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151226 · created 2026-08-06T14:40:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457383.77686 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.029
Transit depth (Rp/Rs)^2	1.15 +/- 0.61 [%]
Orbital Inclination (inc)	86.77 +/- 2.83
Airmass coefficient 1 (a1)	1.175 +/- 0.022
Airmass coefficient 2 (a2)	-0.153 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.91 %
Best Comparison Star	None
Optimal Aperture	3.48
Optimal Annulus	16.35
Transit Duration (day)	0.127 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.029 vs 0.1178 (deviation 0.29σ)
Duration fit vs published	0.127 d vs 0.125 d (deviation 0.34σ)
Mid-time O–C	1571.2 min
Depth SNR	2.8
Residual scatter	1.91 %

Light curve

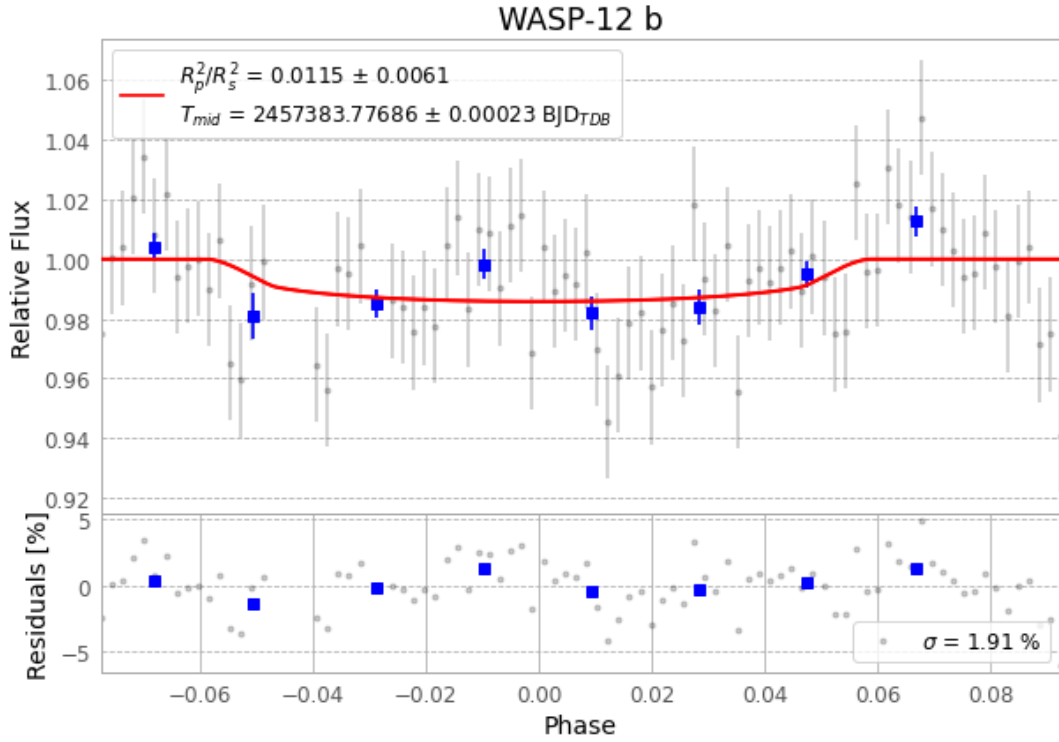


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-26.png (SHA-256 adb7e3ae5eebf909...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b24e8f10c18cc79e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 651d290

Data provenance

88 FITS frames (58 MB), WASP-12151227042712.FITS → WASP-12151227085412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151226.log (bbfc4fa0cd17...), FinalLightCurve_WASP-12 b_2015-12-26.png (adb7e3ae5eeb...), FinalLightCurve_WASP-12 b_2015-12-26.csv (042d86d35bda...)

Compute resources

Wall time 161.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:40Z.